

Direct & Inverse Proportion

Write the $\propto$, find k , write the link, then substitute. Every proportion question is the same four moves: L-I-N-K.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Direct & Inverse Proportion

01 Proportion in words

cost of n items

workers and days

02 The LINK method

$$y = kx$$

$$y = \frac{k}{x}$$

03 Squares, cubes and roots

$$M = kh^3$$

$$y = \frac{k}{\sqrt{x}}$$

04 Graphs and tables

line through O

read a grid point

05 Two links

$$y \propto \frac{1}{\sqrt{x}}, x \propto T^3$$

06 Percentage change

$$1.15^2$$

$$2^3$$

01 Proportion in Words

No letters, no k ? Then it is a words question. Ask one thing first: **more of one — more or less of the other?**

WHEN YOU SEE → YOU DO

“in direct proportion”, cost of n items, recipe

→ find **one** unit, then multiply

“workers / machines / people ... days / hours”

→ inverse: find the **total work** first

“assume they work at the same rate”

→ comment: if slower, it takes longer

“paid for each hour or part of an hour”

→ round the hours **up**

Direct

more items → more cost divide, then multiply

Inverse

more workers → fewer days multiply (total work), then divide

1

Direct: find one

5 pens cost £3.40 → 1 pen = $3.40 \div 5 = £0.68$ → 8 pens = £5.44.

2

Inverse: find the total work

workers $\times$ days = total work (worker-days). Then divide by the new number of workers.

WORKED MODEL • MACHINES

6 machines make all the boxes in 9 days. Days 1, 2, 3 only 3, 4, 5 machines work; then all 6. Total days?

1 $6 \times 9 = 54$ machine-days

total work

2 $3 + 4 + 5 = 12$ done in 3 days

first days

3 $(54 - 12) \div 6 = 7$ more days

the rest

ANSWER $3 + 7 = 10$ days

× **TRAP** Inverse is **not** “divide the days”. More workers → fewer days: always go through the total work.

IDEA 01

Practise: Proportion in Words

My questions first, then selected practice.

Q1 DR ESLAM Find one

CH 1 • TOPIC 15 2 marks

5 pens cost £3.40. Work out the cost of 8 of these pens.

Q2 DR ESLAM Total work

CH 1 • TOPIC 15 2 marks

6 workers build a wall in 10 days. How many days would 4 workers take, working at the same rate?

SELECTED PRACTICE

Q3**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 3 marks

- 1 Yesterday it took 5 cleaners $4\frac{1}{2}$ hours to clean all the rooms in a hotel.

There are only 3 cleaners to clean all the rooms in the hotel today.

Each cleaner is paid £8.20 for each hour or part of an hour they work.

How much will each cleaner be paid today?

(3 marks)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 4 marks

9 (a) To complete a task in 15 days a company needs

4 people each working for 8 hours per day.

The company decides to have

5 people each working for 6 hours per day.

Assume that each person works at the same rate.

How many days will the task take to complete?

You **must** show your working.

(3 marks)

(b) Comment on how the assumption affects your answer to part (a).

(1 mark)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 8 marks

- 14 (a)** Sam and two friends put letters in envelopes on Monday. The three of them take two hours to put 600 letters in envelopes.

On Tuesday Sam has three friends helping.

Working at the same rate, how many letters should the **four** of them be able to put in envelopes in two hours?

(2 marks)

- (b)** Working at the same rate, how much longer would it take **four** people to put 1000 letters in envelopes than it would take **five** people?

(4 marks)

- (c)** Sam says

It took two hours for three people to put 600 letters in envelopes.

If I assume they work all day, then in one day three people will put 7200 letters in envelopes because $600 \times 12 = 7200$.

Why is Sam's assumption not reasonable? What effect has Sam's assumption had on her answer?

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15

6 marks

15 (a) Donald swims 3 lengths of a swimming pool in 93 seconds.

i) Use this information to show that he could swim 100 lengths in under 55 minutes.

[4]

ii) What assumption did you make in part (i)?

[1]

(5 marks)

(b) Donald tries to swim the 100 lengths in under 55 minutes.

Suggest one reason why he might not achieve this.

(1 mark)

Q7**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 3 marks

15 15 machines work at the same rate. Together, the 15 machines can complete an order in 8 hours.

3 of the machines break down after working for 6 hours. The other machines carry on working until the order is complete.

In total, how many hours does **each** of the other machines work?

.....hours

(3 marks)

Q8**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15

4 marks

- 11** Beth and Mia translate documents from Spanish into English.
A set of documents that would take Beth 8 days would take Mia 10 days.

Beth starts to translate the documents.

After 2 days Beth and Mia both work on translating the documents.

How many **more** days will it take to complete the work?

You **must** show your working.

.....days

(4 marks)

02 The LINK Method: Direct and Inverse

Every algebra proportion question is solved with the same four steps. Learn the name: **L-I-N-K**.

WHEN YOU SEE → YOU DO

“y is (directly) proportional to x”,
“varies as”

$$\rightarrow y = kx$$

“y is inversely proportional to x”

$$\rightarrow y = \frac{k}{x}$$

“find a formula / equation linking”

→ stop after step 3

“find the value of ...”

→ do all 4 steps

1

L — Layout

Write $\propto$, then the equation with k : $y \propto \frac{1}{x} \Rightarrow y = \frac{k}{x}$.

2

I — Insert

Put the **first pair** of numbers in to find k . Never round k .

3

N — New equation

Rewrite with the value of k . This is the linking formula.

4

K — Keep solving

Substitute the new value and solve.

WORKED MODEL • INVERSE

y is inversely proportional to x . $y = 36$ when $x = 1.5$. Find y when $x = 6$.

1 $y = \frac{k}{x}$ L

2 $36 = \frac{k}{1.5} \Rightarrow k = 54$ I

3 $y = \frac{54}{x}$ N

4 $y = \frac{54}{6}$ K

ANSWER $y = 9$

CHECK Direct: x up $\rightarrow y$ up. Inverse: x up $\rightarrow y$ down. If not, look again at step 1.

IDEA 02

Practise: The LINK Method: Direct and Inverse

My questions first, then selected practice.

Q9 DR ESLAM Direct

CH 1 • TOPIC 15 3 marks

y is directly proportional to x . $y = 20$ when $x = 4$. Find y when $x = 7$.

Q10 DR ESLAM Inverse

CH 1 • TOPIC 15 3 marks

p is inversely proportional to q . $p = 6$ when $q = 4$. Find p when $q = 3$.

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Medium

CH 1 • TOPIC 15 3 marks

2 p is inversely proportional to t .

When $t = 4$, $p = 12$

Find the value of p when $t = 6$

(3 marks)

Q12 **PRACTICE** Practice · Medium

CH 1 • TOPIC 15 3 marks

3 d is inversely proportional to c

When $c = 280$, $d = 25$

Find the value of d when $c = 350$

(3 marks)

Q13 **PRACTICE** Practice · Medium

CH 1 • TOPIC 15 3 marks

4 y is inversely proportional to x

When $x = 1.5$, $y = 36$

Find the value of y when $x = 6$

(3 marks)

Q14 **PRACTICE** Practice · Medium

CH 1 • TOPIC 15 3 marks

12 y is inversely proportional to x .
 $y = 0.04$ when $x = 80$.

Find the value of y when $x = 32$.

 $y = \dots\dots\dots$

(3 marks)

Q15**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 1 mark

- 16 $V = \frac{k}{H}$ where k is a constant. Which **two** statements are correct?

Tick **two** boxes.

☐	V is directly proportional to H
☐	V is inversely proportional to H
☐	V is directly proportional to $\frac{1}{H}$
☐	V is inversely proportional to $\frac{1}{H}$

(1 mark)

Q16**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 1 mark

17 $xy = c$ where c is a constant. Circle the correct statement.

y is directly proportional to x	y is directly proportional to $\frac{1}{x}$
y is inversely proportional to $\frac{1}{x}$	x is directly proportional to y

(1 mark)

03 Squares, Cubes and Roots

Same four steps. The only new thing: the power sits on x in step 1 — and you undo it at the end.

WHEN YOU SEE → YOU DO

“the square of d ”, “ d^2 ”

$$\longrightarrow y = kd^2 \text{ or } y = \frac{k}{d^2}$$

“varies as the cube of h ”

$$\longrightarrow M = kh^3 \rightarrow \text{at the end take } \sqrt[3]{}$$

“the square root of q ”, “ $\sqrt{q}$ ”

$$\longrightarrow y = k\sqrt{q} \text{ or } y = \frac{k}{\sqrt{q}}$$

“the positive value of x ”

$$\longrightarrow \text{square root} \rightarrow \text{take the } + \text{ root}$$

directly proportional to the square of x

$$y = kx^2$$

inversely proportional to the cube of x

$$y = \frac{k}{x^3}$$

directly proportional to the square root of x

$$y = k\sqrt{x}$$

inversely proportional to $\sqrt{x}$

$$y = \frac{k}{\sqrt{x}}$$

WORKED MODEL • JUNE 2022 P2H

M varies directly as the cube of h . $M = 4$ when $h = 0.5$. Find h when $M = 500$.

1 $M = kh^3$

L

2 $4 = k(0.5)^3 = 0.125k \Rightarrow k = 32$

I

3 $M = 32h^3$

N

4 $500 = 32h^3 \Rightarrow h^3 = 15.625$

K

ANSWER $h = \sqrt[3]{15.625} = 2.5$

× **TRAP** Put the power on x **before** you find k : $4 = k \times 0.5$ is wrong — it is $4 = k \times 0.5^3$.

IDEA 03

Practise: Squares, Cubes and Roots

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Square

CH 1 • TOPIC 15 3 marks

y is directly proportional to the square of x . $y = 50$ when $x = 5$. Find y when $x = 6$.

Q18 DR ESLAM Inverse square

CH 1 • TOPIC 15 3 marks

y is inversely proportional to x^2 . $y = 3$ when $x = 2$. Find y when $x = 6$.

SELECTED PRACTICE

Q19**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 4 marks

6 (a) T is directly proportional to the cube of r

$$T = 21.76 \text{ when } r = 4$$

Find a formula for T in terms of r

(3 marks)

(b) Work out the value of T when $r = 6$

(1 mark)

Q20**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 3 marks

4 The intensity of the sound, I watts/m², received from a loudspeaker is inversely proportional to the square of the distance, d metres, from the loudspeaker.

When $d = 2$, $I = 30$ Work out the value of I when $d = 10$

(3 marks)

Q21**PRACTICE**

Practice · Hard

CH 1 • TOPIC 15 5 marks

14 (a) y is inversely proportional to $\sqrt{x}$

$$y = 4 \quad \text{when} \quad x = 9$$

Work out an equation connecting y and x .

(3 marks)

(b) Work out the value of y when $x = 25$

(2 marks)

Q22**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15 5 marks

- 9 (a) y is directly proportional to the cube of x

$$y = 20h \text{ when } x = h \quad (h \neq 0)$$

Find a formula for y in terms of x and h

$$y = \dots\dots\dots$$

(3 marks)

- (b) Find x in terms of h when $y = 67.5h$

Give your answer in its simplest form.

$$x = \dots\dots\dots$$

(2 marks)

PAST PAPERS

Q23**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q14

CH 1 • TOPIC 15**3 marks**

14 F is inversely proportional to the square of v .

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v .

(Total for Question 14 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 2H • Q16**CH 1 • TOPIC 15** **6 marks**

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

(b) Find the value of A when $r = 7.5A$

$$A = \dots\dots\dots$$

(Total for Question 16 is 6 marks)

Q25 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 1H • Q15**CH 1 • TOPIC 15** **3 marks**

15 A is inversely proportional to C^2

$$A = 40 \text{ when } C = 1.5$$

Calculate the value of C when $A = 1000$

$$C = \dots\dots\dots$$

(Total for Question 15 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q17

CH 1 • TOPIC 15 4 marks

17 M varies directly as the cube of h

$$M = 4 \text{ when } h = 0.5$$

Find the value of h when $M = 500$

(Total for Question 17 is 4 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q17

CH 1 • TOPIC 15 4 marks

17 F is inversely proportional to the square of r

$$F = 36 \text{ when } r = 4$$

(a) Find a formula for F in terms of r

(b) Work out the value of F when $r = 48$

(Total for Question 17 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q17

CH 1 • TOPIC 15 3 marks

17 Q is directly proportional to the square root of d

$$Q = 4.5 \text{ when } d = 324$$

Find a formula for Q in terms of d

(Total for Question 17 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q14

CH 1 • TOPIC 15 3 marks

14 B is inversely proportional to the square of d

$$B = 0.25 \text{ when } d = 12$$

Find a formula for B in terms of d

(Total for Question 14 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q18

CH 1 • TOPIC 15

5 marks

18 F is inversely proportional to the cube of r

$$F = 6 \text{ when } r = 2$$

(a) Find a formula for F in terms of r

.....

(b) Find the value of r when $F = 3072$

$$r = \text{.....}$$

(Total for Question 18 is 5 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q18

CH 1 • TOPIC 15

4 marks

18 y is inversely proportional to x^3

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \text{.....}$$

(Total for Question 18 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q17

CH 1 • TOPIC 15 3 marks

17 T is inversely proportional to $\sqrt{m}$

$$T = 15 \text{ when } m = 36$$

Find a formula for T in terms of m

.....
(Total for Question 17 is 3 marks)

04 Graphs and Tables

Know the four shapes by heart. A table? Test $\frac{y}{x}$ (direct) or xy (inverse) — if it stays the same, you found the link.

WHEN YOU SEE → YOU DO

straight line through the origin

$$\rightarrow y \propto x$$

curve going down, never touching the axes

$$\rightarrow y \propto \frac{1}{x}$$

“the graph shows ... find a formula”

→ read one clear point, use it as the first pair

“complete the table” / “show the values fit”

→ find k from the full column, then use it

$$y \propto x$$

straight line through O

a line *not* through O
is not proportion

$$y \propto x^2$$

curve bending **up** from O

gets steeper

$$y \propto \sqrt{x}$$

curve bending **over** from O

gets flatter

$$y \propto \frac{1}{x}, \frac{1}{x^2}$$

curve coming down towards the x -axis

never meets the axes

WORKED MODEL • JUNE 2023 P2H

Q is directly proportional to $\sqrt{t}$. The graph passes through $(4, 6)$. Find Q in terms of t .

1 $Q = k\sqrt{t}$

L

2 $6 = k\sqrt{4} = 2k \Rightarrow k = 3$

I • pick a grid point

ANSWER $Q = 3\sqrt{t}$

PICK THE POINT Choose a point where the curve crosses a grid corner exactly — and where $\sqrt{t}$ is nice ($t = 1, 4, 9$).

IDEA 04

Practise: Graphs and Tables

My questions first, then selected practice, then real past papers.

Q33 **DR ESLAM** Test the table

CH 1 • TOPIC 15 2 marks

x : 2, 4, 8 y : 12, 6, 3.

Is y directly or inversely proportional to x ? Give the formula.

SELECTED PRACTICE

Q34 **PRACTICE** Practice • Medium

CH 1 • TOPIC 15 2 marks

7 y is inversely proportional to x .

Complete the table.

x	12	6	
y		4	8

(2 marks)

Q35**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 4 marks

- 3 Given that $y \propto \frac{1}{x^2}$, complete this table of values.

x	1	2	5	10
y				1

(4 marks)

Q36**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 5 marks

- 12 (a) The following table gives values of x and y where y is inversely proportional to the square of x .

x	1.5	2	3	4
y	16	9	4	2.25

Find a formula for y in terms of x .

(3 marks)

- (b) Given that $x > 0$

find the value of x when $y = 144$

(2 marks)

Q37

PRACTICE

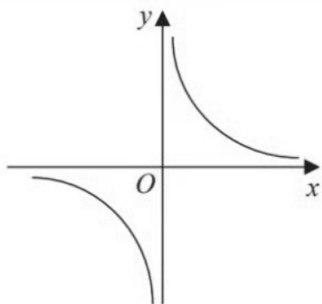
Practice • Very Hard

CH 1 • TOPIC 15

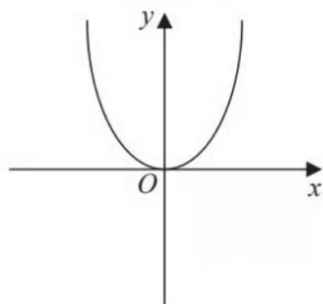
2 marks

PRACTISE • 1.15

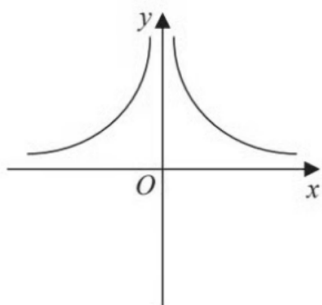
6 These graphs show four different proportionality relationships between y and x .



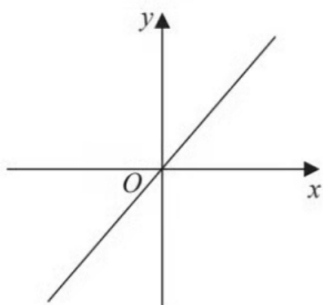
Graph A



Graph B



Graph C



Graph D

Match each graph with a statement in the table below.

Proportionality relationship	Graph letter
y is directly proportional to x	
y is inversely proportional to x	
y is proportional to the square of x	
y is inversely proportional to the square of x	

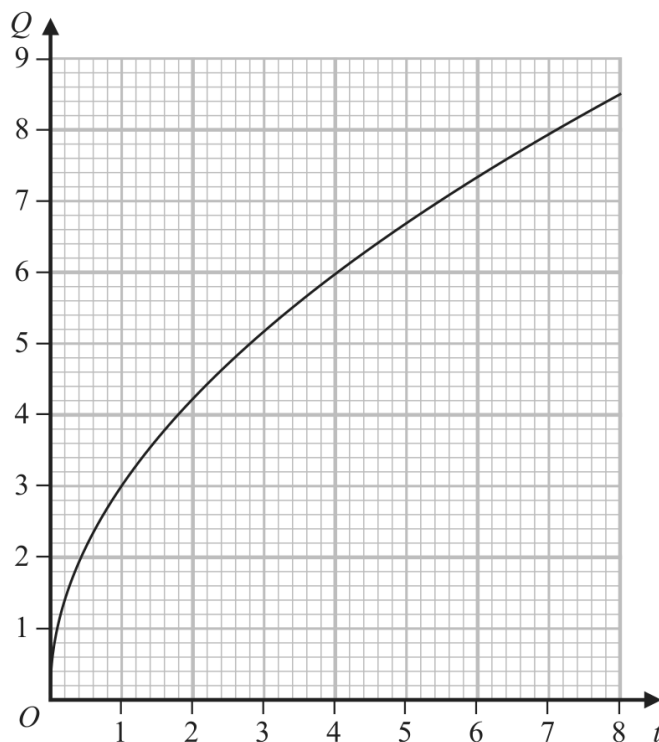
(2 marks)

PAST PAPERS

Q38**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q16

CH 1 • TOPIC 15 5 marks

16 Q is directly proportional to $\sqrt{t}$ The graph shows the relationship between Q and t for $0 < t < 8$ (a) Find a formula for Q in terms of t

.....

 Q is increased by 20%(b) Find the percentage increase in t

.....%

(Total for Question 16 is 5 marks)

05 Chained Proportion (Two Links)

Two sentences with $\propto$? Do LINK twice, then put Link 2 inside Link 1. Nothing new.

WHEN YOU SEE → YOU DO

" $y \propto \dots d$ and $d \propto \dots x$ "	→ Link 1 ($y-d$), Link 2 ($d-x$)
"find a formula for y in terms of x "	→ substitute Link 2 into Link 1
only one pair given for both links	→ use one constant K for the chain
"exact value"	→ keep fractions and roots — no decimals

1

Link 1

Find k for the first pair of letters.

2

Link 2

Find the second constant for the other pair.

3

Substitute

Replace the middle letter, then simplify. The middle letter must disappear.

WORKED MODEL • JANUARY 2022 P2HR

$y \propto \frac{1}{\sqrt{x}}$, $x \propto T^3$. $y = 8$ when $T = 25$. Find the exact value of T when $y = 27$.

$$1 \quad y = \frac{k}{\sqrt{cT^3}} = \frac{K}{T^{3/2}}$$

one chain constant

$$2 \quad 8 = \frac{K}{25^{3/2}} = \frac{K}{125} \Rightarrow K = 1000$$

I

$$3 \quad 27 = \frac{1000}{T^{3/2}} \Rightarrow T^{3/2} = \frac{1000}{27}$$

K

$$\text{ANSWER} \quad T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

× **TRAP** Do not use the numbers of Link 1 inside Link 2. Each link has its own pair of values.

IDEA 05

Practise: Chained Proportion (Two Links)

My questions first, then selected practice, then real past papers.

Q39**DR ESLAM**

Two links

CH 1 • TOPIC 15

4 marks

y is directly proportional to z ; $y = 12$ when $z = 4$. z is inversely proportional to x ; $z = 3$ when $x = 2$. Find a formula for y in terms of x .

SELECTED PRACTICE

Q40 PRACTICE Practice • Very Hard

CH 1 • TOPIC 15 5 marks

- 3 y is inversely proportional to d^2
When $d = 10$, $y = 4$

d is directly proportional to x^2
When $x = 2$, $d = 24$

Find a formula for y in terms of x .
Give your answer in its simplest form.

(5 marks)

Q41 **PRACTICE** Practice • Very Hard

CH 1 • TOPIC 15 4 marks

7 h is inversely proportional to p

p is directly proportional to $\sqrt{t}$

Given that $h = 10$ and $t = 144$ when $p = 6$
find a formula for h in terms of t

(4 marks)

Q42**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15 5 marks

13 P , Q and R have positive values.

P is directly proportional to the square of Q .

When $P = 1.25$, $Q = 0.5$

Q is inversely proportional to R .

When $Q = 0.5$, $R = 6$

Work out the value of R when $P = 0.8$

(5 marks)

Q43**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15

4 marks

- 16** x is directly proportional to y .
 y is directly proportional to z .

When $x = 10$, $y = 60$.

When $y = 8$, $z = 1.6$.

Find a formula for z in terms of x .

(4 marks)

PAST PAPERS

Q44 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 2HR • Q20**CH 1 • TOPIC 15** **4 marks**

- 20** y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

06 Percentage Change: Use Multipliers

No numbers for k ? You do not need k . Turn the percentage into a multiplier and push it through the power.

WHEN YOU SEE → YOU DO

“ x is increased by 15%”

→ multiply by 1.15

“is doubled / multiplied by 4”

→ put 2 or 4 in for x

“find the percentage increase in y ”

→ $(\text{multiplier} - 1) \times 100$

“ Q is increased by 20%, find the % increase in t ”

→ work backwards: undo the power

$$y \propto x^2, x \times 1.15$$

$$y \times 1.15^2 = 1.3225$$

up 32.25%

$$D \propto n^3, n \times 2$$

$$D \times 2^3 = 8$$

not $\times 6$

$$m \propto \frac{1}{\sqrt{r}}, r \times 4$$

$$m \times \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$m \div 2$

$$V \propto \frac{1}{p}, V \times 1.25$$

$$p \times \frac{1}{1.25} = 0.8$$

down 20%

WORKED MODEL • JUNE 2023 P2H

$Q = 3\sqrt{t}$. Q is increased by 20%. Find the percentage increase in t .

1 $t = \left(\frac{Q}{3}\right)^2$, so $t \propto Q^2$

make t the subject

2 $t \times 1.2^2 = t \times 1.44$

multiplier through the power

ANSWER t increases by 44%

× **TRAP** A 20% increase in Q is **not** a 20% increase in t — the power changes the percentage.

IDEA 06

Practise: Percentage Change: Use Multipliers

My questions first, then selected practice.

Q45 DR ESLAM Multiplier

CH 1 • TOPIC 15 3 marks

y is directly proportional to x^3 . x is decreased by 10%. Find the percentage decrease in y .

SELECTED PRACTICE**Q46** PRACTICE Practice • Very Hard

CH 1 • TOPIC 15 3 marks

- 4 A pendulum of length L cm has time period T seconds.
 T is directly proportional to the square root of L .

The length of the pendulum is increased by 40%.

Work out the percentage increase in the time period.

(3 marks)

Q47**PRACTICE**

Practice · Very Hard

CH 1 • TOPIC 15 **1 mark**

- 5 D is directly proportional to the cube of n .

Mary says that when n is doubled, the value of D is multiplied by 6

Mary is wrong.

Explain why.

(1 mark)

Q48**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 15 4 marks

12 (a) y is directly proportional to x^3

$$y = 17 \text{ when } x = 4$$

Work out an equation connecting y and x .

(3 marks)

(b) m is inversely proportional to $\sqrt{r}$.The value of r is multiplied by 4.Circle what happens to the value of m .

$\times 2$	$\times 16$	$\div 2$	$\div 16$
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(1 mark)

Q49**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 15 4 marks

14 y is directly proportional to the square of x .

Find the percentage increase in y when x is increased by 15%.

(4 marks)**Q50****PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 15 4 marks

15 At a constant temperature, the volume of a gas V is inversely proportional to its pressure p .
By what percentage will the pressure of a gas change if its volume increases by 25%?

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------------|--|
| 01 | Words | Direct: find one, then multiply. Inverse: total work first, then divide. |
| 02 | LINK | Layout ($y = kx$ or $\frac{k}{x}$) → Insert (find k) → New equation → Keep solving. |
| 03 | Powers & roots | Power on x <i>before</i> you find k ; undo it at the end. |
| 04 | Graphs & tables | Line through O = direct. Test $\frac{y}{x}$ or xy . Read an exact grid point. |
| 05 | Two links | LINK twice, then substitute; the middle letter disappears. |
| 06 | Percentages | Multiplier through the power: 1.15^2 , 2^3 , $\frac{1}{\sqrt{4}}$. |

BOTTOM LINE Layout → Insert (find k) → New equation → Keep solving.
 “Formula” = stop at N. “Value” = go to K.

Answers

Direct & Inverse Proportion · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PROPORTION IN WORDS

- | | | |
|-----------|-----------------------|--|
| Q1 | Dr Eslam · Find one | £5.44 |
| Q2 | Dr Eslam · Total work | $6 \times 10 = 60$ worker-days; $60 \div 4 = 15$ days |
| Q3 | Practice · Medium | 22.5 cleaner-hours $\div 3 = 7.5$ h $\rightarrow$ paid 8 h:
£65.60 |
| Q4 | Practice · Medium | (a) 480 person-hours $\div 30 = 16$ days (b) if
some work slower it takes longer |
| Q5 | Practice · Medium | (a) 800 (b) $2.5 - 2 = 0.5$ h = 30 min longer
(c) they cannot keep the same rate all day;
overestimate |
| Q6 | Practice · Medium | (a)(i) 31 s per length; 3100 s = 51 min 40 s
< 55 min (ii) same speed every length (b) he
gets tired |
| Q7 | Practice · Hard | 8.5 hours |
| Q8 | Practice · Very Hard | $3\frac{1}{3}$ more days |

IDEA 02 • THE LINK METHOD: DIRECT AND INVERSE

Q9 Dr Eslam • Direct

$$y = 5x; y = 35$$

Q10 Dr Eslam • Inverse

$$p = \frac{24}{q}; p = 8$$

Q11 Practice • Medium

$$p = \frac{48}{t}; p = 8$$

Q12 Practice • Medium

$$d = \frac{7000}{c}; d = 20$$

Q13 Practice • Medium

$$y = \frac{54}{x}; y = 9$$

Q14 Practice • Medium

$$y = \frac{3.2}{x}; y = 0.1$$

Q15 Practice • Hard

V is inversely proportional to H ; V is directly proportional to $\frac{1}{H}$

Q16 Practice • Hard

y is directly proportional to $\frac{1}{x}$

IDEA 03 • SQUARES, CUBES AND ROOTS

- Q17** Dr Eslam • Square $y = 2x^2; y = 72$
- Q18** Dr Eslam • Inverse square $y = \frac{12}{x^2}; y = \frac{1}{3}$
- Q19** Practice • Medium (a) $T = 0.34r^3$ (b) 73.44
- Q20** Practice • Hard $I = \frac{120}{d^2}; I = 1.2$
- Q21** Practice • Hard (a) $y = \frac{12}{\sqrt{x}}$ (b) 2.4
- Q22** Practice • Very Hard (a) $y = \frac{20x^3}{h^2}$ (b) $x = \frac{3}{2}h$
- Q23** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q14 $F = \frac{104}{v^2}$
- Q24** Edexcel 4MA1 • May 2021 • Paper 2H • Q16 (a) $A = \frac{0.45}{r^2}$ (b) $A^3 = 0.008, A = 0.2$
- Q25** Edexcel 4MA1 • January 2022 • Paper 1H • Q15 $A = \frac{90}{C^2}; C = 0.3$
- Q26** Edexcel 4MA1 • June 2022 • Paper 2H • Q17 $M = 32h^3; h = 2.5$
- Q27** Edexcel 4MA1 • May 2023 • Paper 1HR • Q17 (a) $F = \frac{576}{r^2}$ (b) 0.25
- Q28** Edexcel 4MA1 • June 2024 • Paper 2H • Q17 $Q = 0.25\sqrt{d}$
- Q29** Edexcel 4MA1 • November 2024 • Paper 1H • Q14 $B = \frac{d^2}{576}$
- Q30** Edexcel 4MA1 • June 2025 • Paper 1HR • Q18 (a) $F = \frac{48}{r^3}$ (b) $r = 0.25$
- Q31** Edexcel 4MA1 • June 2025 • Paper 1H • Q18 $y = \frac{108}{x^3}; x = 0.5$
- Q32** Edexcel 4MA1 • November 2025 • Paper 2H • Q17 $T = \frac{90}{\sqrt{m}}$

IDEA 04 • GRAPHS AND TABLES

- Q33** Dr Eslam • Test the table $xy = 24$ every time: inverse, $y = \frac{24}{x}$
- Q34** Practice • Medium $x: 12, 6, 3 \cdot y: 2, 4, 8$
- Q35** Practice • Hard $y = \frac{100}{x^2}: 100, 25, 4, 1$
- Q36** Practice • Hard (a) $y = \frac{36}{x^2}$ (b) $x = 0.5$
- Q37** Practice • Very Hard $y \propto x: D \cdot \frac{1}{x}: A \cdot x^2: B \cdot \frac{1}{x^2}: C$
- Q38** Edexcel 4MA1 • June 2023 • Paper 2H • Q16 (a) $Q = 3\sqrt{t}$ (b) 44%

IDEA 05 • CHAINED PROPORTION (TWO LINKS)

- Q39** Dr Eslam • Two links $y = 3z, z = \frac{6}{x} \Rightarrow y = \frac{18}{x}$
- Q40** Practice • Very Hard $y = \frac{400}{d^2}, d = 6x^2 \Rightarrow y = \frac{100}{9x^4}$
- Q41** Practice • Very Hard $h = \frac{60}{p}, p = 0.5\sqrt{t} \Rightarrow h = \frac{120}{\sqrt{t}}$
- Q42** Practice • Very Hard $P = 5Q^2, Q = \frac{3}{R}; Q = 0.4, R = 7.5$
- Q43** Practice • Very Hard $y = 6x, y = 5z \Rightarrow z = 1.2x$
- Q44** Edexcel 4MA1 • January 2022 • Paper 2HR • Q20 $y = \frac{1000}{T^{3/2}}; T = \frac{100}{9}$

IDEA 06 • PERCENTAGE CHANGE: USE MULTIPLIERS

Q45 Dr Eslam • Multiplier

$$0.9^3 = 0.729: y \text{ decreases by } 27.1\%$$

Q46 Practice • Very Hard

$$\sqrt{1.4} = 1.183: 18.3\%$$

Q47 Practice • Very Hard

$$2^3 = 8: D \text{ is multiplied by } 8, \text{ not } 6$$

Q48 Practice • Very Hard

$$(a) y = \frac{17}{64}x^3 \quad (b) \div 2$$

Q49 Practice • Very Hard

$$1.15^2 = 1.3225: 32.25\%$$

Q50 Practice • Very Hard

$$\frac{1}{1.25} = 0.8: \text{pressure decreases by } 20\%$$